320612(20)

BE (6th Semester)
Examination, April-May, 2016

(Old Scheme)

Geotech Engineering-II

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Time Allowed : 3 hours *Maximum Marks* : 80
Minimum Pass Marks : 28

Note : (*i*) Part (*a*) of each question is compulsory. Attempt any **two** parts of (*b*), (*c*) and (*d*).

(*ii*) The figures in the right-hand margin indicate marks.

1. (*a*) What is the importance of slope stability analysis? 2

(*b*) Explain Swedish slip circle method. 7

(*c*) A slope is to be constructed at an inclination of 30° with the horizontal. Determine the safe height of the slope at factor of safety of 1.5. the soil has the following properties : $c = 15$ kN/m^2, $\phi = 22.5°$ and $\gamma = 19$ kN/m^3. 7

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(*d*) A new canal is excavated to a depth of 5m below ground level, through a soil having the following characteristics– $c = 14$ kN/m^2, $\phi = 15°$, $e = 0.8$ and $G = 2.70$. The slope of banks is 1 in 1. Calculate the factor of safety with respect to cohesion when the canal runs full. If it is suddenly and completely emptied, what will be the factor of safety? 7

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2. (*a*) What do you mean by state of general plastic equilibrium? 2

(*b*) Explain briefly the active and passive earth pressure. 7

(*c*) Compute the intensities of active and passive earth pressure at depth of 8m in dry cohesionless sand with an angle of interval friction of 30° and unit weight of 18 kN/m^3. What will be the intensities of active and passive earth pressure if the water level rises to the ground level? Take saturated unit weight of sand as 22 kN/m^3. 7

(*d*) A retaining wall, 4m high supports a backfill [$c = 20$ kN/m^2, $\phi = 30°$, $\gamma = 20$ kN/m^3] with horizontal top, flush with the top of the wall. The backfill carries a surcharge of 20 kN/m^2. If the wall is pushed towards the backfill, compute the total passive pressure on the wall, and its point of application. 7

3. (*a*) What do you mean by bearing capacity of soil? 2

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(*b*) Discuss the effect of water table on the bearing capacity of soil. 7

(*c*) A square footing 1.2m × 1.2m rests at a depth of 1m in a saturated clay layer 4m deep. the clay is normally consolidated, having an unconfined compressive strength of 40 kN/m^2. CSVTUonline.com
The soil has a liquid limit of 30%, γ_{sat}=17.8 kN/m^3, ω = 28% and G=2.68. Determine the load which the footing can carry safely with a factor of safety of 3 against shear. Also, determine the settlement if the footing is loaded with this safe load. Use Terzaghis analysis for bearing capacity. 7

(*d*) Define following : 7

(*i*) Footing
(*ii*) Bearing capacity
(*iii*) Foundation
(*iv*) Ultimate Bearing capacity
(*v*) Net ultimate Bearing Capacity
(*vi*) Gross pressure intensity
(*vii*) Safe Bearing Capacity

4. (*a*) Where pile foundation must be provided? Give the criteria for pile foundation. 2

(*b*) Discuss the various forces acting on a well foundation. CSVTUonline.com 7

(*c*) In a 16 pile group, the pile diameter is 45 cm and centre to centre spacing of the square group is 1.5 m. If c = 50 kN/m^2, determine whether the failure would occur with the pile acting individually, or as a group. Neglect bearing at the tip of the pile. All piles are 10 m long. Take m = 0.7 for shear mobilisation around each pile.

(*d*) Design of friction pile group to carry a load of 3000 kN including the weight of the pile cap at a site where the soil is uniform clay to a depth of 20 m, underlain by rock. Average unconfined compressive strength of the clay is 70 kN/m^2. The clay may be assumed to be of normal sensitivity and normally loaded, with liquid limit 60%. A factor of safety of 3 is required against shear failure. 7
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5. (*a*) What are CNS soils? 2

(*b*) Briefly explain the Barkan method of machine foundation design. 7

(*c*) Using Barken's expressions for natural frequency and the amplitude of vibrations, calculate the change in the percentage amplitude in terms of γ if the soil mass participating in the vibrations is 23% of m. Also, calculate this change for γ = 0.3 and γ = 2. CSVTUonline.com 7

(*d*) Write short notes on (any three) : 7

(*i*) Block vibration test
(*ii*) Design criteria for dynamic type of machines
(*iii*) Bulb of pressure concept
(*iv*) IS code method for impact type of machines

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